

Pillar 5: Capstone Project

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Post Graduate Diploma in Artificial Intelligence and Machine Learning

PHASE 08 | FetalGuard AI (CTG — Fetal Risk Classifier) — Deployment Guide & MLOps Practices

Important Notes

- Use files in folder '02 src' for demonstration

Step 1 — Get the prerequisite file

Phase 08 needs `ctg_preprocessed.pkl` from Phase 02. Copy it into the `ctg_mlops` folder:

```
cp ctg_preprocessed.pkl ctg_mlops/
```

Step 2 — Install dependencies

```
cd ctg_mlops
pip install -r requirements.txt
```

Step 3 — Build the deployment artifact

This trains the model and bundles everything into one file:

```
python ../streamlit_app/build_artifact.py
```

You should see:

```
Training best model (XGBoost + scale_pos_weight)...
Top 5 features: ['ASTV', 'AC', 'Variance', 'ALTV', 'Width']
Saved deployment_artifact.pkl
```

Step 4 — Start the API

```
uvicorn api.main:app --host 0.0.0.0 --port 8000 --reload
```

You'll see:

```
INFO ctg_api Loading model artifact ...
INFO ctg_api Model loaded: ctg-fetal-risk-classifier v1.0.0 |
features=22 | threshold=0.10
INFO Uvicorn running on http://0.0.0.0:8000
```

Step 5 — Use it

Option A — Interactive browser UI (easiest)

Open `http://localhost:8000/docs` — this is FastAPI's built-in Swagger interface. You can click any endpoint, fill in values, and click Execute.

Option B — curl from terminal

Health check

```
curl http://localhost:8000/health
```

Single prediction (top-5 features – abnormal example)

```
curl -X POST http://localhost:8000/predict \
  -H "Content-Type: application/json" \
  -d '{"ASTV": 80, "AC": 0, "Variance": 60, "ALTV": 70, "Width": 20}'
```

Single prediction (normal example)

```
curl -X POST http://localhost:8000/predict \
  -H "Content-Type: application/json" \
  -d '{"ASTV": 30, "AC": 5, "Variance": 2, "ALTV": 0, "Width": 80}'
```

Full 22-feature prediction

```
curl -X POST http://localhost:8000/predict/full \
  -H "Content-Type: application/json" \
  -d '{"LB": 120, "ASTV": 80, "AC": 0, "Variance": 60}'
```

Batch (multiple patients at once)

```
curl -X POST http://localhost:8000/predict/batch \
  -H "Content-Type: application/json" \
  -d '{"inputs": [{"ASTV": 80, "AC": 0}, {"ASTV": 30, "AC": 5}]}'
```

Override the threshold (Lower = more sensitive)

```
curl -X POST http://localhost:8000/predict \
  -H "Content-Type: application/json" \
  -d '{"ASTV": 40, "AC": 0, "Variance": 10, "threshold": 0.10}'
```

Running stats

```
curl http://localhost:8000/metrics
```

Option C — Python

```
import requests
```

```
response = requests.post(
    "http://localhost:8000/predict",
    json={"ASTV": 80, "AC": 0, "Variance": 60, "ALTV": 70, "Width": 20}
```

```
)  
print(response.json())  
# {'prediction': 'High Risk', 'probability': 0.999167, 'threshold':  
0.10, 'model_version': '1.0.0'}
```

Step 6 — Run the tests (optional but recommended)

```
pytest tests/test_api.py -v  
All 29 should pass.
```

Step 7 — Track experiments with Weights & Biases (optional)

```
# Log in (first time only)  
wandb login  
  
# Log a training run  
python wandb_track.py  
  
# Open the W&B dashboard  
# The run URL is printed in the terminal after wandb_track.py finishes,  
e.g.:  
# https://wandb.ai/<your-entity>/ctg-mlops/runs/<run-id>
```

Step 8 — Run in Docker (optional)

```
# Build the image  
docker build -t ctg-api:1.0.0 .  
  
# Run it  
docker run -p 8000:8000 ctg-api:1.0.0  
  
# Test it (same curl commands as above still work)  
curl http://localhost:8000/health
```